

§2. Def. transformării Laplace. Example.

Cursul 11
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Funcție original ($\mathcal{O}$)

$$- f(t) = 0; \forall t < 0$$

- f

- f - adică exp. $\sigma_0 = \sigma_0(f) =$ decizia conv.

$$\exists M \geq 0: |f(t)| \leq M \cdot e^{at}; \forall t \in \mathbb{R}.$$

1. Def $f \in \mathcal{O}$

$$\Delta_0 = \Delta(f) = \{s \in \mathbb{C}: \operatorname{Re} s > \sigma_0\}$$

$$\text{Funcția } \bar{f}: \Delta_0 \rightarrow \mathbb{K}; \bar{f}(s) = \int_0^{\infty} f(t) e^{-st} dt \quad \forall s \in \Delta_0$$

se numește transformata Laplace a originalului f .

$$\text{Not } \bar{f}(s) = \mathcal{L}\{f(t)\}(s)$$

$\mathcal{L}$ - operator Laplace
 $\bar{f} = \mathcal{L} \cdot f$

2. Def Procedul prin care se face asocierea $f \in \mathcal{O} \rightarrow \bar{f} = \mathcal{L}f$ se numește transformare Laplace

3. Obs

Integrala care definește transformata Laplace este uniform și absolut convergentă pentru $s \in \Delta_0$ și $f \in \mathcal{O}$.

Funcția $\bar{f}(s)$ - domeniul în semiplanul Δ_0 și derivată se poate obține prin derivare sub semnul integral.

$$F'(s) = \int_0^{\infty} f(t) \cdot (-t) e^{-st} dt$$

$$= - \int_0^{\infty} t f(t) e^{-st} dt$$

$$= - \mathcal{L}\{t f(t)\}(s)$$

$$= \mathcal{L}\{-t f(t)\}(s)$$

$$F^{(n)}(s) = \mathcal{L}\{(-t)^n f(t)\}(s) = (-1)^n \mathcal{L}\{t^n f(t)\}(s)$$

4. Example:

$$4.1. \mathcal{L}\{e^{at}\}(s) = \frac{1}{s-a}; \operatorname{Re} s > \frac{\operatorname{Re} a}{\sigma_0}$$

Demonstration:

$$\mathcal{L}\{e^{at}\}(s) = \int_0^{\infty} e^{at} e^{-st} dt = \int_0^{\infty} e^{(a-s)t} dt$$

$$= \frac{1}{a-s} e^{(a-s)t} \Big|_0^{\infty}$$

$$= \frac{1}{a-s} \left[\lim_{t \rightarrow \infty} e^{(a-s)t} - 1 \right] \quad (1)$$

$$a = a_1 + ja_2$$

$$s = s_1 + js_2$$

$$e^{(a-s)t} = e^{(a_1-s_1)t} e^{j(a_2-s_2)t}$$

$$= e^{(a_1-s_1)t} \cdot e^{j(a_2-s_2)t} \quad (2)$$

$$|e^{j(a_2-s_2)t}| = 1$$

$$\lim_{t \rightarrow \infty} e^{(a_1-s_1)t} = \begin{cases} e^{-\infty} = 0; & a_1 - s_1 < 0 \\ e^0 = 1; & a_1 - s_1 = 0 \\ e^{\infty} = \infty; & a_1 - s_1 > 0 \end{cases}$$

$$(2) \Rightarrow \lim_{t \rightarrow \infty} e^{(a_1 - b_1)t} \cdot e^{f(a_2 - b_2)t} = 0$$

$$\text{dacă } b_1 > a_1 \Leftrightarrow \operatorname{Re} b > \operatorname{Re} a \quad (3)$$

$$\text{Pentru } a_1 = a_2 \Rightarrow \lim_{t \rightarrow \infty} e^{f(a_2 - b_2)t}$$

$$\Rightarrow \lim_{t \rightarrow \infty} e^{(a - b)t}$$

În concluzie, transformata Laplace a originalului $f(t) = e^{at}$ este definită, dacă $\operatorname{Re} b > \operatorname{Re} a$.

conform (1) și (3)

$$\mathcal{L}\{e^{at}\}(s) = \frac{1}{s-a} \quad (\operatorname{Re} s > \operatorname{Re} a)$$

$$4.2 \quad \mathcal{L}\{t^x\}(s) = \frac{\Gamma(x+1)}{s^{x+1}}; \quad x > -1; \operatorname{Re} s > 0$$

Demonstrație

$$\mathcal{L}\{t^x\}(s) = \int_0^{\infty} t^x e^{-st} dt$$

$$\Gamma(a) = \int_0^{\infty} x^{a-1} e^{-x} dx; \quad a > 0$$

$$st = x \Rightarrow t = \frac{1}{s} \cdot x \Rightarrow dt = \frac{1}{s} dx$$

$$\mathcal{L}\{t^x\}(s) = \int_0^{\infty} \left(\frac{1}{s} x\right)^x \cdot e^{-x} \cdot \frac{1}{s} dx$$

$$= \frac{1}{s^{x+1}} \int_0^{\infty} x^x e^{-x} dx$$

$$\stackrel{a-1=x}{\stackrel{a=x+1}}{=} \frac{1}{s^{x+1}} \cdot \Gamma(x+1)$$

$$4.3. \quad \mathcal{L}\{t^n f(s)\} = \frac{n!}{s^{n+1}}; n \in \mathbb{N} \quad \text{at} \quad e^{at} = e^{at} \cdot u(t) = \begin{cases} 0, & t < 0 \\ e^{at}, & t > 0 \end{cases}$$

$$\Gamma(n+1) = n!$$

$$\mathcal{L}\{u(t)\}(s) = \mathcal{L}\{1\}(s)$$

$$\mathcal{L}\{t\}(s) = \mathcal{L}\{t \cdot u(t)\}(s) = \frac{1}{s^2}$$

$$\mathcal{L}\left\{\frac{1}{\sqrt{t}}\right\}(s) \stackrel{\alpha = -\frac{1}{2}}{=} \frac{1}{s^{\frac{1}{2}}}$$

$$\Gamma\left(-\frac{1}{2} + 1\right) = \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

$$\Gamma(a) \Gamma(1-a) = \frac{\pi}{\sin(\pi a)}$$

§3. Proprietăți ale transformării Laplace

3.1. Liniaritatea

$$\mathcal{L}\{\alpha f(t) + \beta g(t)\}(s) = \alpha \mathcal{L}\{f(t)\}(s) + \beta \mathcal{L}\{g(t)\}(s)$$

$\alpha, \beta \in \mathbb{C}$
 $f, g \in \mathcal{O}$

Example:

$$\begin{aligned} \mathcal{L}\{\sin at\}(s) &= \mathcal{L}\left\{\frac{1}{2j} (e^{jat} - e^{-jat})\right\}(s) \\ &= \frac{1}{2j} [\mathcal{L}\{e^{jat}\}(s) - \mathcal{L}\{e^{-jat}\}(s)] \end{aligned}$$

$$\stackrel{\substack{\text{4.5} \\ a = ja \text{ sau } -ja}}{=} \frac{1}{2j} \left[\frac{1}{s - ja} - \frac{1}{s + ja} \right]$$

$$= \frac{1}{2j} \cdot \frac{s + ja - s + ja}{s^2 + a^2} = \frac{1}{2j} \cdot \frac{2ja}{s^2 + a^2} = \frac{a}{s^2 + a^2}$$

$$\mathcal{L}\{\sin at\}(s) = \frac{a}{s^2 + a^2}; a \in \mathbb{K}; \operatorname{Re} s > 0$$

$$(i) \mathcal{L}\{\cos at\}(s) = \frac{s}{s^2 + a^2}; a \in \mathbb{K}; \operatorname{Re} s > 0$$

$$(ii) \mathcal{L}\{\sinh at\}(s) = \frac{a}{s^2 - a^2}; a \in \mathbb{K}; \operatorname{Re} s > |a|$$

$$(iii) \mathcal{L}\{\cosh at\}(s) = \frac{s}{s^2 - a^2}; a \in \mathbb{K}; \operatorname{Re} s > |a|$$

3.2 Teorema asemănării (schimbare de scală sau comprimare timpului)

$a > 0$

$$\mathcal{L}\{f(at)\}(s) = \frac{1}{a} F\left(\frac{s}{a}\right) = \frac{1}{a} \mathcal{L}\{f(t)\}\left(\frac{s}{a}\right)$$

3.3 Teorema întârzierii originalului

$a > 0$

$$\mathcal{L}\{f(t-a)\}(s) = e^{-as} F(s) = e^{-as} \mathcal{L}\{f(t)\}(s)$$

↳ factor de întârziere

3.4. Teorema deplasării imaginilor

$a \in \mathbb{K}$

$$\mathcal{L}\{e^{at} f(t)\}(s) = F(s-a) = \mathcal{L}\{f(t)\}(s-a)$$

3.5. Teorema derivării originalului

(i) $n=1$

$$f, f' \in \mathcal{O} \Rightarrow \mathcal{L}\{f'(t)\}(s) = sF(s) - f(0+0)$$

$$= s \cdot \mathcal{L}\{f(t)\}(s) - \lim_{t \rightarrow 0} f(t)$$

(ii) $n \geq 1$

$$f^{(k)} \in \mathcal{O} ; 0 \leq k \leq n$$

$$\Rightarrow \mathcal{L}\{f^{(n)}(t)\}(s) = s^n F(s) - [s^{n-1}f(0+0) + s^{n-2}f'(0+0) + \dots + f^{(n-1)}(0+0)]$$

$$= s^n \mathcal{L}\{f(t)\}(s) - \sum_{k=0}^{n-1} s^{n-k-1} \cdot f^{(k)}(0+0)$$

Demonstration (pt casuel $n=1$) - ! pt Examens !

$$\mathcal{L}\{f(t)\}(s) = \int_0^{\infty} f(t) \cdot e^{-st} dt$$

$$= f(t) \cdot e^{-st} \Big|_0^{\infty} - \int_0^{\infty} f(t) (e^{-st})'_t dt$$

$$= \lim_{t \rightarrow \infty} f(t) e^{-st} - \lim_{t \rightarrow 0} f(t) + s \int_0^{\infty} f(t) \cdot e^{-st} dt$$

$$= \lim_{t \rightarrow \infty} f(t) \cdot e^{-st} - f(0+0) + s \mathcal{L}\{f(t)\}(s) \quad (1)$$

$$f \in \mathcal{O} \Rightarrow \exists M \geq 0 : |f(t)| \leq M e^{+at}$$

$$|f(t) e^{-st}| = |f(t)| \cdot |e^{-(\sigma_1 + j\sigma_2)t}|$$

$$= |f(t)| \cdot |e^{-\sigma_1 t}| \cdot |e^{-j\sigma_2 t}|$$

$$= |f(t)| \cdot e^{-\sigma_1 t}$$

$$\Rightarrow |f(t)| e^{-\sigma_0 t} \leq M e^{\sigma_0 t} e^{-\sigma_0 t} = M e^{(\sigma_0 - \sigma_0)t} = M e^{-\sigma_0 t} \quad (1)$$

dar $f \in \mathcal{O} \Rightarrow \operatorname{Re} \lambda \geq \sigma_0 \Rightarrow \lambda_1 > \sigma_0 \Rightarrow \sigma_0 - \lambda_1 < 0$

$$\Rightarrow \lim_{t \rightarrow \infty} e^{(\sigma_0 - \lambda_1)t} = e^{-\infty} = 0 \quad (3)$$

Ann (2)

$$\Rightarrow \lim_{t \rightarrow \infty} f(t) e^{-\sigma_0 t} = 0$$

(3)

$$\Rightarrow \mathcal{L}\{f'(t)\}(\lambda) = \lambda \mathcal{L}\{f(t)\}(\lambda) - f(0+0)$$

$$(11) \quad \mathcal{L}\{f'(t)\}(\lambda) = \lambda \mathcal{L}\{f(t)\}(\lambda) - f(0+0)$$

$$f = f'$$

$$\Rightarrow \mathcal{L}\{f''(t)\}(\lambda) = \lambda \mathcal{L}\{f'(t)\}(\lambda) - f'(0+0)$$

$$f = f^{(n)}$$

$$\Rightarrow \mathcal{L}\{f^{(n)}(t)\}(\lambda) = \lambda \mathcal{L}\{f^{(n-1)}(t)\}(\lambda) - f^{(n-1)}(0+0)$$

$$\mathcal{L}\{f^{(n)}(t)\}(\lambda) = \lambda^n \mathcal{L}\{f(t)\}(\lambda) - [\lambda^{n-1} f(0+0) + \lambda^{n-2} f'(0+0) + \dots + f^{(n-1)}(0+0)]$$

3.6. Teorema derivării imaginii

$$\mathcal{L}\{t^n \cdot f(t)\}(\lambda) = (-1)^n \cdot [\mathcal{L}\{f(t)\}(\lambda)]^{(n)}$$

$$[\mathcal{L}\{f(t)\}(\lambda)]^{(n)} = \mathcal{L}\{(-t)^n \cdot f(t)\}(\lambda)$$

3.7 Teorema integrării originale

$$f \in \mathcal{O} \Rightarrow \mathcal{L}\left\{\int_0^t f(x) dx\right\}(s) = \frac{1}{s} \mathcal{L}\{f(t)\}(s) = \frac{F(s)}{s}$$

3.8 Teorema integrării imaginii

Dacă $f \in \mathcal{O}$ și $\frac{f(t)}{t} \in \mathcal{O}$, atunci

$$\begin{aligned} \mathcal{L}\left\{\frac{f(t)}{t}\right\}(s) &= \int_0^\infty F(y) dy \\ &= \int_0^\infty \mathcal{L}\{f(t)\}(y) dy \quad ; \operatorname{Re} s > \sigma_0\left(\frac{f(t)}{t}\right) \end{aligned}$$

3.9 Corolar

$$\begin{aligned} F(s) = \mathcal{L}\{f(t)\}(s) \\ \frac{f(t)}{t} \in \mathcal{O} \end{aligned} \quad \Rightarrow \quad \int_0^\infty \frac{f(t)}{t} dt = \int_0^\infty F(s) ds$$

3.10 Transformata Laplace a funcțiilor periodice

Dacă $f \in \mathcal{O}$ și $T > 0$; $f(t+T) = f(t) \forall t \geq 0$.

$$\Rightarrow \mathcal{L}\{f(t)\}(s) = \frac{1}{1 - e^{-sT}} \int_0^T f(t) \cdot e^{-st} dt$$

3.11 Dezvoltarea în serie a originalei

$$f \in \mathcal{O}; f(t) = \sum_{n=0}^{\infty} a_n t^n; t \geq 0$$

$$\mathcal{L}\{f(t)\}(s) = \mathcal{L}\left\{\sum_{n=0}^{\infty} a_n t^n\right\}(s) = \sum_{n=0}^{\infty} a_n \cdot \frac{n!}{s^{n+1}}; \operatorname{Re} s > 0$$

3.12 Comportarea la ∞ a transformatei

$$f \in \mathcal{O} \text{ și } F(s) = \mathcal{L}\{f(t)\}(s)$$

$$\Rightarrow \lim_{\operatorname{Re} s \rightarrow \infty} F(s) = 0$$

3.13. Teorema valorii inițiale
 Dacă $f, f' \in \mathcal{O}$ și $\lim_{s \rightarrow \infty} s \cdot F(s) = f(0+0)$

3.14. Teorema valorii finale
 $f, f' \in \mathcal{O}$; f' - mărginită și $f(\infty) = \lim_{t \rightarrow \infty} f(t)$, atunci

$$\lim_{s \rightarrow 0} s \cdot F(s) = f(\infty)$$

3.15 Teorema produsului de convoluție pt transformarea Laplace

(i) Def $f, g \in \mathcal{O}$

$$(f * g)(t) = \int_{-\infty}^t f(x) \cdot g(t-x) dx$$

$$(f * g)(t) = \int_0^t f(x) \cdot g(t-x) dx$$

$$(ii) f, g \in \mathcal{O} \text{ și } F(s) = \mathcal{L}\{f(t)\}(s) \\ G(s) = \mathcal{L}\{g(t)\}(s)$$

$$\Rightarrow \mathcal{L}\{(f * g)(t)\}(s) = F(s) \cdot G(s)$$

$$= \mathcal{L}\{f(t)\}(s) \cdot \mathcal{L}\{g(t)\}(s)$$

3.16. Formula lui Duhamel

Dacă $f, g \in \mathcal{O}$ și g - derivabilă, atunci:

$$s \cdot F(s) \cdot G(s) = \mathcal{L}\{$$

Demonstrație:

$$(f * g)(t) = \int_0^t f(x) \cdot g(t-x) dx = \mathcal{L}\{f(t) \cdot g(0) + \int_0^t f(x) \cdot g'(t-x) dx\}(s) \\ \xrightarrow{(*)} (f * g)'(t) = t' f(t) \cdot g(t-t) - g' + \int_0^t f(x) \cdot g'(t-x) dx$$

$$= f(t) \cdot g(0) + \int_0^t f(x) \cdot g'(t-x) dx \quad | \quad \mathcal{L}$$

$$\mathcal{L}\{(f * g)(t)\}(s) = \mathcal{L}\left\{f(t) \cdot g(0) + \int_0^t f(x) g'(t-x) dx\right\}(s)$$

$$\stackrel{T.3(ii)}{\Rightarrow} \Delta \mathcal{L}\{(f * g)(t)\}(s) - (f * g)(0) = -11-$$

$$\stackrel{T.15(ii)}{\Rightarrow}$$

$$\Delta F(s) \cdot G(s) = -11-$$